

Handoid: Inter-morphologic Robotic Hand Avatar for Multi Presence

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Figure 1: Handoid is a hand-shaped interface. (A) In independent mode, Handoid extends its reach by walking, (B) In integrated mode, Handoid acts as part of a humanoid telexistence robot, (C) Conceptual overview of Handoid, illustrating its morphology.

Abstract

Robotic hands have been developed by many researchers to emulate human hand dexterity to enhance object manipulation. However, the mainstream of the robotic hand has been designed as part of the user's body, and research scopes have been limited to improving functions to achieve human hands. In this paper, we propose *Handoid*, a novel robotic hand avatar that switches its morphology between acting as a part of a humanoid robot and an independent hand-shaped robot avatar physically separated from the main body. The prototype consists of a human-like robotic hand, an avatar robot acting as a user's alter ego, a detachable mechanism for seamless attachment/detachment, and control software incorporating machine learning for generating walking motions. In the demonstration, users can experience the intuitive control of the telexistence robot, and the semi-autonomous walking of the robotic hand. This opens up novel possibilities for robotic hands.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computer systems organization** → **Robotics**.

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Keywords

telepresence, human robot interaction, hand-shaped interface

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1 Introduction

For several decades, there has been research on robotic hands that emulate human hand movements, with the aim of pursuing the ability to grasp objects [Piazza et al. 2019]. The findings from these researches have been implemented not only in humanoid robots but also in telexistence [Tachi et al. 2020] and telepresence robots, which enable users to operate a robot avatar from a remote location and manipulate its body as a user's alter ego. Furthermore, some research explored applications beyond the general scale and shape of the human body. Transfantom [Izumihara et al. 2019] allows users to move freely between two bodies with two different body scales, and achieve flexible body scale conversion.

However, in the majority of previous research, the robotic hand has always been fixedly designed as part of the user's body, and research scopes have been limited to improving functions to achieve human hands. Only a few research aim to expand the range of applications of interaction by freely attaching and detaching wearable

robotic hands to the body [Yoshida et al. 2023]. In this paper, we propose *Handoid*, a novel robotic hand avatar that functions as both an independent robotic hand avatar and a part of a humanoid robot avatar, expanding beyond the conventional role of a robotic hand as a component of a telepresence robot and enabling multi-presence.

2 Concept - Handoid

Handoid is a hand-shaped robotic avatar that switches its morphology between functioning as part of a humanoid robot avatar and operating as an independent robot avatar, to enhance the embodiment and presence of remote users (see Figure 1c). The former morphology is an integrated mode that functions as a robotic hand with the conventional telepresence robot, and the latter is an independent mode physically separated from the humanoid avatar to work as a robotic hand avatar. Users can select either mode depending on the application need, thereby expanding their operational workspace. In independent mode, both the robotic head of the humanoid avatar and the separated robotic hand provide different viewpoints, enabling multi-presence. Moreover, in independent mode, the robotic hand is controlled by blending the user's input motion and context-driven autonomous motion generated by machine learning, allowing it to act as a shared avatar.

3 Prototype

In order to clarify the concept of *Handoid*, we have developed a prototype system consisting of four key elements: (1) a robotic hand as the core of *Handoid*, (2) an avatar robot serving as a user's alter ego, (3) a detachable mechanism for seamless transition between integrated and independent modes, and (4) a control software that incorporates machine learning for generating walking motions. By integrating these elements, we achieve an experience that goes beyond the limitations of traditional telepresence.

The robotic hand is designed to imitate the human right-hand, as shown in Fig. 2A, and this prototype achieves six degrees of freedom (DoF). Each finger is driven by a single-axis servomotor via a link mechanism to achieve flexion and extension. In addition, a single-axis motor at the base of the thumb enhances human-like hand movement. Furthermore, the robotic hand is equipped with a camera and IMU (see Fig. 2B) to support autonomous operation.

The avatar robot functions as a user's alter ego during remote operation. This prototype has a head and a right arm, each with three DoF. This configuration enables users to look around their

surroundings and adjust the robotic hand's posture. In addition, the detachable mechanism is mounted at the end of the arm.

The detachable mechanism uses a simple and highly reliable method, utilizing magnets and a cam mechanism to easily attach and detach the robotic hand from the avatar robot. Thus, in integrated mode, the robotic hand operates as part of the avatar robot. If necessary, it can switch to independent mode and perform semi-autonomous walking and other actions.

The control software functions as a motion engine that blends the user's hand motion data with walking motion data generated by machine learning [Sasaki et al. 2025]. Since the robotic hand with only six DoF cannot fully cover the high-dimensional motion of the human hand, when operated solely by user, forward movement is still challenging. However, with this control software, users can control the robotic hand's motion intuitively and effectively.

4 User Experience

We believe that *Handoid* can be used in a variety of ways and be effective when utilized in conjunction with a telepresence system that has a manipulator arm with a human-like shape.

At SIGGRAPH '25, attendees can experience *Handoid* as follows:

- (1) **Integrated Mode:** *Handoid* functions as a part of the telepresence robot, operating as a robotic hand.
- (2) **Independent Mode:** *Handoid* is detached from the telepresence robot, thereby assuming the role of a hand-shaped robotic avatar that walks independently.

Attendees first use a VR HMD to enter the telepresence robot. The robot's head and a right-arm move with the head and hand tracking of HMD. In integrated mode, they control *Handoid* to reach, touch or grab nearby objects. To reach out of range objects, they switch *Handoid* to independent mode. The detachable mechanism detaches the robotic hand, which then operates as an independent, hand-shaped avatar. In this way, *Handoid* provides attendees with an extended operating area and a sense of immersion by switching between modes and realizes a new interaction experience that goes beyond conventional telepresence systems.

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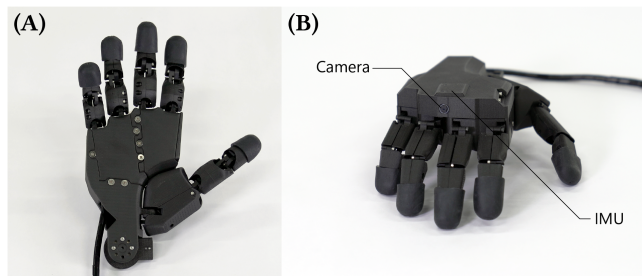


Figure 2: Prototype of Handoid. (A) Robotic hand with six DoF, and (B) Sensor placement (Camera and IMU).